Optimizing Retrieval-Augmented Generation for Querying Locally Stored Public Procurement Documents using Small Language Models

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***Abstract*—This study developed an optimized, locally-deployed Retrieval-Augmented Generation (RAG) system for intelligently querying Philippine public procurement documents. The system processes unstructured PDFs through metadata extraction, OCR-based text extraction, and section-aware chunking (256 512 tokens) that preserves legal document hierarchy. A hybrid retrieval model combines semantic embeddings with hierarchical multi-level filtering across document, section, and chunk levels to maximize retrieval precision. Three Small Language Models— Gemma-3-1B, LLaMA-3.2-1B, and Phi-1.5-1B—were adapted using Quantized Low-Rank Adaptation (QLoRA) with 4-bit quantization, enabling fine-tuning within a 4GB VRAM constraint. Experimental evaluation on 2,088 question-answer pairs from Republic Act No. 9184 procurement regulations demonstrated that Gemma-3-1B achieved statistically significant improvement (p=0.0029), with the optimized RAG system attaining 89.3% overall accuracy compared to 71.3% baseline—an 18% improvement. The complete system operates with a 1.5GB memory footprint, enabling offline deployment on standard government hardware while maintaining data privacy. Expert validation by five procurement officers confirmed practical deployment readiness with an overall rating of 4.38/5.0.**

Keywords*—*Small Language Model, Retrieval-Augmented Generation, Natural Language Processing, Vector Embeddings, Artificial Intelligence

I. Introduction

Public procurement documents, including contracts, bidding records, and compliance reports, are essential for ensuring transparency and accountability in government operations. However, these documents often exist in unstructured PDF formats with complex layouts combining narrative text, legal clauses, and tabular data, making efficient information retrieval challenging for procurement officers and auditors [1].

Recent advances in Retrieval-Augmented Generation (RAG) have shown promising results for document question answering by combining retrieval mechanisms with generative language models [2]. However, existing RAG implementations face significant limitations when applied to procurement document analysis. First, most systems rely on cloud infrastructure and large language models requiring substantial computational resources [3], making them impractical for government institutions with limited hardware capabilities. Second, cloud-based solutions raise data privacy concerns when handling sensitive procurement information [4]. Third, general-purpose language models lack specialized understanding of procurement terminology, legal structures, and compliance requirements defined under regulations such as Republic Act No. 9184 in the Philippines. Small Language Models (SLMs) offer a potential solution due to their reduced computational requirements, but they typically underperform on domain-specific tasks without proper adaptation [5]. Traditional fine-tuning approaches require extensive resources and full model retraining, which remains impractical for resource-constrained environments. Low-Rank Adaptation (LoRA) and its quantized variant QLoRA ad dress this limitation by enabling parameter-efficient fine-tuning (PEFT) through low-rank matrix decomposition, significantly reducing memory requirements while maintaining adaptation effectiveness [6].

To address these challenges, this paper presents an optimized RAG system integrating QLoRA-adapted Small Language Models with a domain-specific retrieval architecture. The proposed approach employs section-aware chunking to preserve document hierarchy, hybrid embedding-classification retrieval for semantic precision, and hierarchical multi-level querying tailored for structured procurement documents.

II. Related Studies

A. RAG Systems

Retrieval-Augmented Generation was introduced by Lewis et al. [3] as a framework combining neural retrieval with sequence generation for knowledge-intensive NLP tasks, demonstrating that grounding language model outputs in retrieved documents significantly reduces hallucination. Izacard and Grave [7] extended this approach through multi-document retrieval, achieving improved performance on open-domain question answering. Gao et al. [2] provided a comprehensive survey of RAG optimization techniques including context pruning and adaptive retrieval strategies.

For efficient retrieval, Khattab et al. [8] proposed Col BERT, achieving significantly faster search times through contextualized late interaction compared to traditional dense retrieval methods. However, ColBERT requires dedicated GPU resources unavailable in many government institutions. Barnett et al. [9] identified seven critical failure points in RAG systems, including embedding mismatches and incomplete context propagation, highlighting the need for domain-specific optimization. Despite these advances, most RAG implementations depend on cloud infrastructure and large language models [2], limiting their applicability for privacy-sensitive government environments.

B. PEFT and Small Language Models

Adapting language models to specialized domains tradition ally requires extensive computational resources. Houlsby et al. [10] introduced adapter modules as a parameter-efficient alternative, inserting small trainable layers into frozen transformer architectures. Hu et al. [6] proposed Low-Rank Adaptation (LoRA), which injects trainable low-rank matrices into trans former layers, enabling domain adaptation without modifying original model weights. Dettmers et al. [12] extended this with QLoRA, combining LoRA with 4-bit quantization to reduce memory requirements below 4GB while maintaining fine-tuning effectiveness.

Recent developments in Small Language Models (SLMs) offer promising alternatives for resource-constrained deployment. Models such as Gemma [13], LLaMA [14], and Phi [15] demonstrate competitive performance with significantly reduced computational requirements. For domain-specific applications, Chalkidis et al. [11] showed that adapted models (LEGAL-BERT) significantly outperform general-purpose models on legal tasks, though their approach requires full pre training rather than parameter-efficient adaptation.

C. Research Gaps

Despite significant progress, several gaps remain unaddressed. First, while Lewis et al. [3] and subsequent RAG systems achieve strong performance, they lack offline deployment capabilities suitable for government environments with limited connectivity and strict data privacy requirements. Second, domain adaptation research has focused on legal [11] and financial documents, leaving the procurement domain unexplored despite its unique challenges including structured tabular data, compliance terminology, and regulatory cross references. Third, no prior work integrates parameter-efficient fine-tuning techniques with domain-specific RAG optimization for locally deployable systems on resource-constrained hard ware.

III. Methodology

A. RAG Architecture

The proposed system implements a three-stage Retrieval Augmented Generation (RAG) architecture comprising Indexing, Retrieval, and Generation pipelines.

Indexing Pipeline: Procurement PDF documents are processed through metadata extraction, OCR-based text extraction for scanned content, and structured table parsing, as shown in Fig. 1. Documents undergo automated section tagging, where structural components are labeled according to procurement document conventions (e.g., [CONTRACT_AMOUNT], [BIDDER_INFO], [LEGAL_CLAUSES]) to preserve document hierarchy. Content is segmented using section-aware chunking (256–512 tokens) that respects semantic boundaries rather than arbitrary character limits. Text and table embeddings are generated using sentence transformers and indexed in a ChromaDB vector database for efficient similarity search.

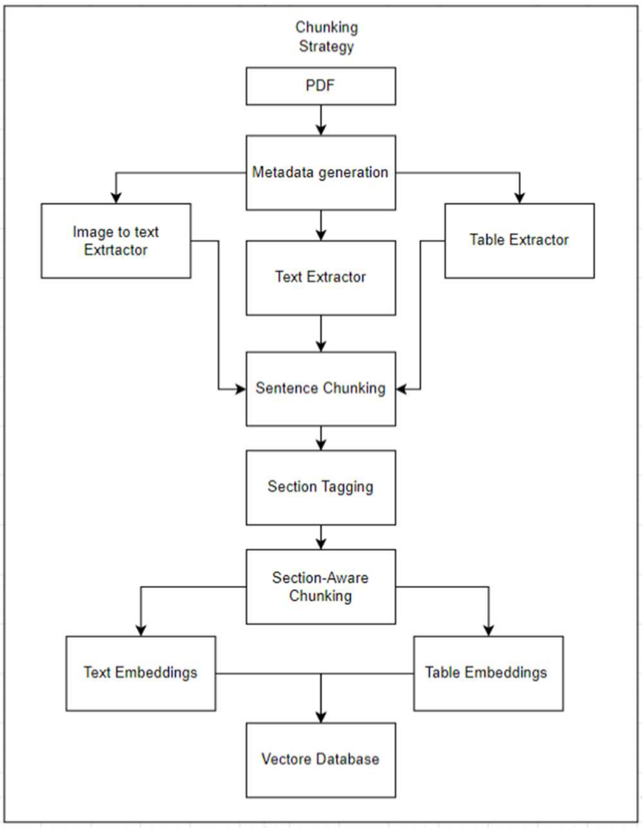

Fig. 1. Indexing Pipeline: Document ingestion through metadata extraction, OCR processing, section tagging, and embedding generation for vector storage.

Retrieval Pipeline: The system employs hierarchical multi-level querying to maximize retrieval precision, as il lustrated in Fig. 2. User queries are encoded into embedding vectors and processed through three filtering levels: (1) document-level filtering identifies relevant source documents, (2) section-level filtering uses structural tags to locate pertinent sections, and (3) chunk-level similarity search retrieves the most contextually relevant passages. Final candidate ranking utilizes a hybrid scoring mechanism:

$$S_{\text{final}} = \alpha Cs + \beta Lj + \gamma Ws + \delta Ms + \epsilon Rmmr \quad (1)$$

where Cs represents cosine similarity, Lj denotes LLM-based relevance judgment, Ws indicates structural weight, Ms captures metadata relevance, and Rmmr applies Maximal Marginal Relevance for diversity.

Weights were optimized through grid search over a held-out validation set of 70 procurement queries with ground-truth annotations sampled from 35 Philippine government bid documents. Table 1 summarizes the search space and optimal configuration. A total of 93 weight combinations satisfied the normalization constraint ($\alpha + \beta + \gamma + \delta + \varepsilon = 1.0$). The optimal configuration achieved MRR@5 = 0.55, with LLM-based judgment receiving the highest weight ($\beta = 0.45$), reflecting the importance of semantic understanding for procurement-specific terminology. Weight optimization was performed once on the validation set and held fixed across all model variants.

TABLE I. RETRIEVAL WEIGHT OPTIMIZATION

Component	Symbol	Search Range	Optimal
Cosine Similarity	α	{0.2, 0.3, 0.35, 0.4, 0.5, 0.6, 0.7}	0.35
LLM Judgment	β	{0.0, 0.2, 0.35, 0.45, 0.5}	0.45
Structural Weight	γ	{0.0, 0.05, 0.1, 0.15}	0.1
Metadata Score	δ	{0.0, 0.05, 0.1}	0.05
MMR Diversity	ϵ	{0.0, 0.05, 0.1}	0.05

The high weight assigned to LLM-based judgment ($\beta = 0.45$) reflects the importance of semantic understanding for procurement-specific terminology. Weight optimization was performed once on the validation set and held fixed across all model variants.

To evaluate the contribution of each scoring component, we conducted an ablation study by systematically removing the LLM-judgment term (β=0) while redistributing its weight to cosine similarity.

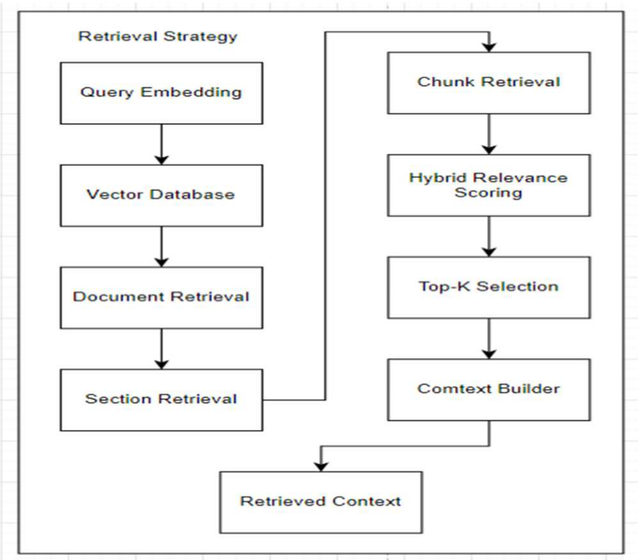

Fig. 2. Retrieval Pipeline: Hierarchical multi-level querying through document, section, and chunk-level filtering with hybrid relevance scoring.

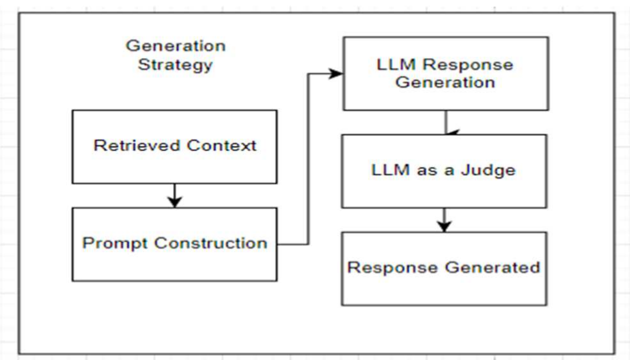

Fig. 3. Generation Pipeline: Context assembly, prompt construction, SLM response generation, and LLM-as-a-Judge validation.

Generation Pipeline: Retrieved context is assembled with user queries into structured prompts, processed by the fine-tuned Small Language Model, as depicted in Fig. 3. An LLM-as-a-Judge mechanism employing the base model validates response faithfulness against retrieved sources before final output delivery, ensuring factual consistency and reducing hallucination.

B. QLoRA Fine-Tuning

Three Small Language Models were selected for domain adaptation based on their balance between performance capability and computational efficiency: Gemma-3-1B, LLaMA 3.2-1B, and Phi-1.5-1B. Table II summarizes model configurations.

TABLE II. QLORA FINE-TUNING CONFIGURATION

Model	Parameters	Quantization	LoRA Rank
Gemma-3-1B	1B	4-bit NF4	8–32
Llama-3.2-1B	1B	4-bit NF4	8
Phi1.5-1B	1B	4-bit NF4	8-16
Model	Parameters	Quantization	LoRA Rank
Gemma-3-1B	1B	4-bit NF4	8–32
Llama-3.2-1B	1B	4-bit NF4	8

Fine-tuning employed Quantized Low-Rank Adaptation (QLoRA) with 4-bit NormalFloat4 (NF4) quantization and bfloat16 compute precision, enabling training within a 4GB VRAM constraint suitable for consumer-grade hardware. The training dataset comprised 2,088 question-answer pairs curated from Philippine procurement regulations under Republic Act No. 9184, encompassing legal clauses, bidding requirements, contract specifications, and compliance documentation. The dataset was partitioned into training (1,775 samples, 85%) and validation (313 samples, 15%) using a fixed seed (seed=42) for reproducibility. To mitigate data leakage, training and test sets were derived from distinct source documents. The training corpus was curated from Government Procurement Policy Board (GPPB) circulars and the Implementing Rules and Regulations (IRR-A), covering general procurement procedures, legal definitions, and compliance requirements. The held-out test set was constructed from actual procurement bid documents (Invitations to Bid, Requests for Quotation) containing institution-specific information such as Approved Budget Costs, submission deadlines, and project specifications. This source-level separation ensures that test queries require extracting document-specific facts rather than memorized regulatory text, reducing evaluation leakage risk. Formal near-duplicate detection was not performed and is acknowledged as a limitation. LoRA adapters were injected into query, key, and value projection layers alongside feed-forward modules within transformer architectures.

C. Evaluation Framework

1. Model Evaluation Metrics: Generation quality was assessed using five metrics: Perplexity (language model confidence), BLEU (n-gram precision against references), ROUGE 1/2/L (recall-based n-gram

overlap), F1 Score (precision-recall harmonic mean), and Semantic Similarity (cosine similarity between output and reference embeddings). Statistical significance of improvements was determined using paired t-tests ($\alpha = 0.05$).

It is important to distinguish two separate uses of LLM-based judgment in this work. First, within the retrieval pipeline, the LLM-judgment component employs the base Gemma-3 1B model to score chunk relevance during reranking, this is an internal system component. Second, for evaluation purposes, an external GPT-4o-mini model serves as an independent evaluator to assess response quality metrics. Using an external model as evaluator prevents self-evaluation bias that would occur if the same model both generated and evaluated responses.

To validate automated evaluation reliability, the authors manually reviewed a random sample of 30 query-response pairs (27% of test set), independently scoring Faithfulness and Answer Correctness using the same rubric provided to GPT-4o-mini. Agreement between human and automated scores reached 90% within ±0.2 tolerance, supporting the use of LLM-as-Judge for large-scale evaluation.

2. All RAG metrics are computed automatically using LLM-as-Judge methodology with GPT-4o-mini (temperature = 0.1). Each metric produces a score in [0, 1].

Faithfulness measures whether factual claims in the generated answer are grounded in retrieved context:

$$\text{Faithfulness}(a, C) = \frac{|\text{claims in } a \text{ supported by } C|}{|\text{claims in } a|} \quad (2)$$

The judge evaluates whether numbers, dates, and facts appear in the top-5 retrieved chunks.

Answer Correctness measures semantic equivalence between generated and ground truth answers:

$$\text{Correctness}(a, a^*) = \text{SemanticSimilarity}(a, a^*) \quad (3)$$

Factual questions require strict matching; descriptive questions evaluate key-point coverage.

Table Context Relevancy evaluates whether retrieved table chunks contain query-relevant data. This metric is applied to queries where the expected answer comes from a table. If no tables are retrieved, TCR = 0.0, penalizing systems that miss relevant tabular information.

Answer Relevancy measures topical alignment between question and response, independent of correctness:

$$AnswerRelevancy = (q, a) = TropicalAlignment(q, a) \quad (4)$$

Evaluation was conducted across 100 queries spanning 50 procurement documents, including both digitally generated and OCR-processed scanned PDFs.

Overall Accuracy is computed as the arithmetic mean of the four RAG metrics:

$$\text{Overall} = \frac{Faithfulness + Correctness + TCR + AR}{4} \quad (5)$$

Each metric is scored in [0, 1], yielding an overall score where ⩾0.80 indicates reliable performance suitable for operational deployment.

3. Expert Validation: System usability and practical applicability were validated by five procurement officers with experience in Republic Act No. 9184 processes, using a 5-point Likert scale across functionality, reliability, usability, and performance dimensions.

4. Evaluation Scope**:** This study employs end-to-end evaluation rather than component ablation studies. Each pipeline component serves a critical retrieval function, and the degradation from removing any component is inherently predictable from its design purpose, the expected negative outcomes are already apparent without formal ablation experiments.

IV. Results and Discussion

A. Model Benchmarking

Table III presents the performance comparison between base and QLoRA fine-tuned models across six evaluation metrics. All three models demonstrated improvements after domain adaptation, with Gemma-3-1B achieving the most substantial gains.

TABLE III. MODEL BENCHMARK RESULTS: BASE VS. FINE-TUNED

Model	PPL	BLEU	R-1	R-L	Sem.	F1
Gemma Base	1617.0	0.016	0.191	0.142	0.545	0.150
Gemma Fine-tuned	5.52	0.062	0.352	0.273	0.609	0.289
LLaMA Base	41.13	0.020	0.175	0.139	0.589	0.195
LLaMA Fine-tuned	25.71	0.020	0.184	0.147	0.604	0.225
Phi Base	30.62	0.021	0.222	0.165	0.415	0.188
Phi Fine-tuned	7.61	0.023	0.215	0.155	0.486	0.224

PPL=Perplexity, R-1=ROUGE-1, R-L=ROUGE-L, Sem.=Semantic Similarity

TABLE IV. STATISTICAL SIGNIFICANCE ANALYSIS (PAIRED T-TEST)

Model	p-value	Significant?
Gemma-3-1B	0.0029	Yes ($p < 0.01$)
LLaMA-3.2-1B	0.0389	Yes ($p < 0.05$)
Phi-1.5-1B	0.1828	No

Gemma-3-1B exhibited a 99.7% reduction in perplexity (1617.0 to 5.52), indicating dramatically improved language fluency and confidence in procurement-specific text generation. Additionally, Gemma achieved the highest gains in BLEU (0.016 to 0.062), ROUGE-1 (0.191 to 0.352), and F1 Score (0.150 to 0.289), confirming superior domain adaptation. The elevated base perplexity reflects the model's lack of exposure to Philippine procurement terminology and legal structures during pre-training, as RA 9184-specific vocabulary constitutes out-of-distribution text for general-purpose models.

Statistical significance was assessed using paired t-tests comparing base and fine-tuned performance across six evaluation metrics (BLEU, ROUGE-1, ROUGE-2, ROUGE-L, Semantic Similarity, F1). To justify the parametric assumption, the Shapiro-Wilk test confirmed normality of difference scores ($W = 0.909$, $p = 0.43$). For Gemma-3-1B, the selected model, fine-tuning produced a highly significant improvement ($t(5) = 5.40$, $p = 0.0029$) with a large effect size (Cohen's $d = 2.20$). Robustness was verified via bootstrap resampling (10,000 iterations, seed = 42), yielding a 95% confidence interval for mean metric improvement of [0.069, 0.136].

Based on these results, Gemma-3-1B was selected as the optimal model for integration into the RAG system.

B. *RAG System Evaluation*

The optimized RAG system was evaluated using 100 queries across 50 procurement documents. Table IV compares the base (non-fine-tuned) and fine-tuned system performance across four metrics.

TABLE V. UNIFIED EVALUATION TABLE

Document Type	Faith.	Corr.	TCR	AR	Overall
Digital PDFs	0.963	0.875	—	0.975	0.938
Scanned/Table PDFs	0.751	0.662	0.9	0.897	0.802
Overall	0.893	0.804	0.9	0.949	0.893

Faith.=Faithfulness, Corr.=Correctness, TCR=Table Context Relevancy, AR=Answer Relevancy

As shown in Table V, digital PDFs achieved 0.938 overall accuracy versus 0.802 for OCR-processed scans, a 14.5% reduction. Faithfulness and Correctness showed the largest degradation (22% and 24% respectively), while Answer Relevancy remained stable (0.975 digital, 0.897 scanned).

OCR accuracy degrades under conditions common in government archives: low scan resolution (<300 DPI), poor contrast, and skewed pages. This asymmetric degradation results from OCR errors propagating through the pipeline: corrupted text yields weak embeddings, reducing retrieval precision. When textually corrupted but topically relevant chunks are retrieved, the model produces responses that address the query intent (high AR: 0.897) but contain factual inconsistencies (low Faithfulness: 0.751, Correctness: 0.662).

TABLE VI. RAG SYSTEM EVALUATION METRICS

Mode	Faith.	Corr.	TCR	AR	Overall
Base	0.704	0.676	0.533	0.940	0.713
Fine-tuned	0.893	0.804	0.900	0.949	0.893

Faith.=Faithfulness, Corr.=Correctness, TCR=Table Context Relevancy, AR=Answer Relevancy

The fine-tuned system achieved an overall accuracy of 89.3%, representing an 18.0% improvement over the 71.3% baseline as shown in Table VI. The most significant improvement occurred in Table Context Relevancy (0.533 to 0.900), demonstrating that domain adaptation substantially enhanced the system's ability to process and integrate tabular procurement data. Fig. 4 illustrates the performance gains across all metrics.

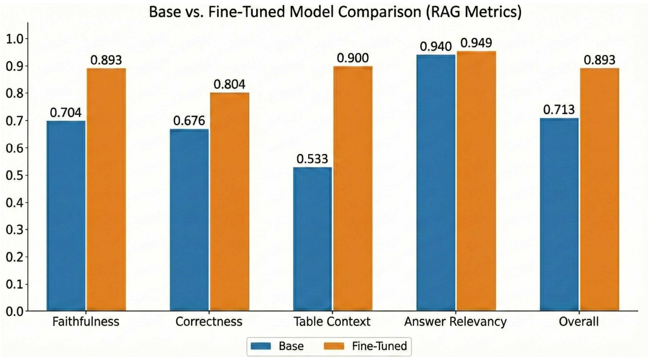

Fig. 4. RAG system performance comparison: Base model vs. Fine-tuned model across four evaluation metrics.

Text Faithfulness improved from 0.704 to 0.893, indicating that generated responses remained more factually consistent with retrieved procurement documents. Answer Correctness increased from 0.676 to 0.804, confirming higher accuracy against validated ground truth answers.

Ablation of LLM-Judgment Reranking**.** To assess the contribution of LLM-based relevance scoring (β), we compared configurations with and without this component. The LLM-judgment uses the base Gemma-3 1B model without domain fine-tuning, serving only to evaluate chunk relevance. With LLM-judgment (β=0.45), MRR@5=0.5455; without (β=0), MRR@5=0.5258, a 3.6% relative reduction. This modest improvement is expected since β functions as a secondary reranking signal, while primary retrieval relies on domain-optimized embeddings. The LLM's more substantial contribution lies in downstream answer generation, where the fine-tuned model synthesizes responses from retrieved context.

C. *Expert Validation and Hardware Performance*

System usability was validated by five procurement officers experienced in Republic Act No. 9184 processes. Expert evaluation yielded positive results across all dimensions: Functionality (4.64/5.0), Reliability (4.50/5.0), Usability (3.98/5.0), and Performance (4.40/5.0), with an overall system rating of 4.38/5.0. Fig. 5 summarizes these results.

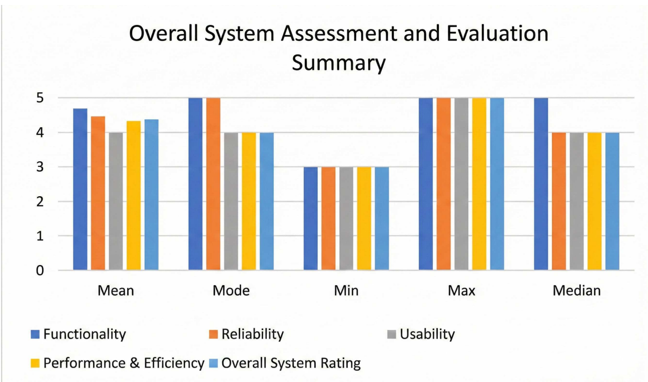

Fig. 5. Expert evaluation scores across four assessment dimensions.

Hardware performance analysis confirmed practical deployment viability across varying system configurations, as shown in Table VII. The complete system operates with a 1.5GB memory footprint, enabling deployment on standard government-issued hardware. Performance analysis revealed that retrieval operations complete in approximately 0.01 seconds, with LLM inference constituting the primary latency factor. These results confirm that the optimized RAG system achieves practical response times while maintaining full offline capability suitable for privacy-sensitive government environments.

TABLE VII. HARDWARE PERFORMANCE BY CONFIGURATION

Configuration	**RAM**	**Response Time**
Low-end	4GB	25–35s
Mid-range	8GB	12–18s
High-end	16GB	7–9s

V. CONCLUSION

This paper addressed three key gaps in RAG system research: the lack of offline deployment capabilities, unexplored procurement domain, and absence of parameter efficient fine-tuning integration with domain-specific retrieval. The proposed architecture combines QLoRA-adapted Small Language Models with section-aware chunking, hierarchical multi-level retrieval, and hybrid relevance scoring specifically designed for structured procurement documents.

Experimental evaluation demonstrated that QLoRA finetuning effectively enables domain adaptation under resource constraints. Gemma-3-1B achieved statistically significant improvements (p=0.0029), outperforming LLaMA-3.2-1B and Phi-1.5-1B across all metrics. The optimized RAG system attained 89.3% overall accuracy compared to 71.3% baseline, with Table Context Relevancy exhibiting the most substantial improvement. These gains validate that domain-specific finetuning is essential for processing structured procurement data containing legal clauses and tabular information.

The system operates fully offline with a 1.5GB memory footprint, enabling deployment on standard government hardware without cloud dependency. Expert validation confirmed practical readiness with functionality and reliability scores exceeding 4.5/5.0. However, limitations remain: OCR accuracy degrades with low-quality scanned documents, and Phi-1.5-1B did not achieve statistically significant adaptation.

This study intentionally restricts its scope to RA 9184 (Government Procurement Reform Act and its amendments like RA 12009) rather than pursuing multi-domain coverage. Domain-specific fine-tuning ensures the model develops specialized comprehension of procurement terminology, document structures, and compliance requirements, a depth of adaptation that would be diluted by training across heterogeneous legal frameworks. The methodology itself remains transferable; however, validating cross-domain performance falls outside the current research objectives.

Future work will explore support for non-PDF document formats common in procurement workflows, integration of multi-language capabilities for cross-border procurement analysis, and continuous adaptation mechanisms to accommodate evolving regulatory frameworks such as amendments to Republic Act No. 9184.

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